



Federated learning meets remote sensing

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ABSTRACT

Remote sensing (RS) imagery provides invaluable insights into characterizing the Earth's land surface within the scope of Earth observation (EO). Technological advances in capture instrumentation, coupled with the rise in the number of EO missions aimed at data acquisition, have significantly increased the volume of accessible RS data. This abundance of information has alleviated the challenge of insufficient training samples, a common issue in the application of machine learning (ML) techniques. In this context, crowd-sourced data play a crucial role in gathering diverse information from multiple sources, resulting in heterogeneous datasets that enable applications to harness a more comprehensive spatial coverage of the surface. However, the sensitive nature of RS data requires ensuring the privacy of the complete collection. Consequently, federated learning (FL) emerges as a privacy-preserving solution, allowing collaborators to combine such information from decentralized private data collections to build efficient global models. This paper explores the convergence between the FL and RS domains, specifically in developing data classifiers. To this aim, an extensive set of experiments is conducted to analyze the properties and performance of novel FL methodologies. The main emphasis is on evaluating the influence of such heterogeneous and disjoint data among collaborating clients. Moreover, scalability is evaluated for a growing number of clients, and resilience is assessed against Byzantine attacks. Finally, the work concludes with future directions and serves as the opening of a new research avenue for developing efficient RS applications under the FL paradigm. The source code is publicly available at <https://github.com/hpc-unex/FLmeetsRS>.

1. Introduction

The rapid evolution of machine learning (ML) and deep learning (DL) technologies has empowered the importance of data science in multiple fields (Carvalho et al., 2019; Falk et al., 2020; Le Glaz et al., 2021; Nassif, Shahin, Attili, Azzeh, & Shaalan, 2019). In particular, remote sensing (RS) is a cutting-edge field of study with widespread applications in Earth observation (EO). The scientific community devotes significant efforts to this active field, given its valuable and positive impact on understanding and predicting the phenomena on the Earth's surface. The beneficial impact of RS applications is noteworthy for a wide range of socioeconomic activities, which range from environmental observation and conservation to resource exploitation, from security and intelligence applications to disaster monitoring and risk prevention (Louw et al., 2022; Poursanidis & Chrysoulakis, 2017; Taloor, Goswami, Bahuguna, Singh, & Kothiyari, 2022), among others.

In this context, image processing techniques are essential to extract useful information from the collected RS data and to provide the desired service to end-users. Extracting and exploiting this information using ML methods offers notable advantages compared to traditional techniques, e.g., in classification applications (O'Mahony et al., 2020). Nevertheless, these methods suffer from several limitations due to their internal characteristics, mode of operation, or the inherent RS data characteristics. In particular, the lack of suitable and sufficient data to adjust the models causes performance degradation. To overcome this drawback, several strategies have been proposed, e.g., oversampling methods aim to address imbalanced datasets by increasing the representation of the minority classes. Also, Generative Adversarial Networks (GANs) generate synthetic data samples that mirror the original distribution. Regularization techniques, such as L1/L2 regularization, impose

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Nomenclature

Most relevant abbreviations in alphabetical order	
CNN	Convolutional Neural Network
CPU	Central Processing Unit
DL	Deep Learning
DNN	Deep Neural Network
EO	Earth Observation
FL	Federated Learning
GPU	Graphic Processing Unit
HPC	High-Performance Computing
LULC	Land-Use Land-Cover
ML	Machine Learning
MPI	Message Passing Interface
ResNet	Residual Network
RS	Remote Sensing
SGD	Stochastic Gradient Descent
UAV	Unmanned Aerial Vehicle

constraints on model parameters during training, thus alleviating overfitting in scenarios with limited data. Dropout randomly deactivates neurons during training to enhance model robustness, among others.

These efforts to artificially increase the number of samples have proven inadequate for real-world applications. In the following, different models will be briefly reviewed for optical image analysis, highlighting their potential and shortcomings, with emphasis on the aforementioned negative impact of the lack of labeled data. Subsequently, insights into crowd-sourced data will be provided, and the federated learning (FL) paradigm will be introduced as a natural solution for RS data processing.

1.1. Optical image analysis: potentials and challenges

Optical RS images characterize the Earth's surface materials by recording the solar spectrum through imaging sensors placed on aerial, orbital, or unmanned aerial vehicle (UAV) platforms. Usually, these instruments sense electromagnetic radiation throughout the vacuum ultraviolet (UV), the visible near-infrared (VNIR), and the solar-reflected short-wavelength infrared (SWIR) wavelengths. The resulting data exhibit specific spatial, spectral, radiometric, and temporal properties, providing a wide range of information that effectively and descriptively represents the composition of the soil surface. This valuable knowledge has found extensive application in a wide range of activities, such as environmental stress in crops and forests analysis (Virnodkar, Pachghare, Patil, & Jha, 2020), species migrations, pests and infestations monitoring (El-Ghany, Nesreen, El-Aziz, Shadia, & Marei, 2020), mineral and geological treatments (Hecker et al., 2019), maritime studies (Reggianini et al., 2019), military defense solutions (Li, Garshick et al., 2021), and catastrophe monitoring and risk prediction (Kaur, Rishi, Singh, & Thakur, 2020; Zhu et al., 2020).

Information is extracted by manifold processing strategies, ranging from specific analyses (e.g., spectral unmixing, pansharpening, image fusion, semantic segmentation, denoising or super-resolution), to more holistic and aggregated assessments (e.g., land-use and land-cover — LULC — classification, few-shot classification, or change, target, and anomaly detection, among others), each tailored to extract specific information from abundant data scenes (Xie et al., 2023). Nevertheless, land-cover analysis with optical RS imaging poses several challenges. Indeed, geographically recorded data capture specific regions, resulting in certain materials being rare or absent in disjoint scenes, thereby contributing to increased content variety. As a consequence, RS scenes may contain a large mixture of land-cover elements due to the low spatial resolution or large scene extent. The challenge of limited spectral

information complicates the accurate distinction of elements exhibiting similar features, which is further exacerbated when transferring features for application across global regions. In addition, long satellite revisit times and different flight campaigns cause the same scene taken at two different times to differ greatly. These drawbacks translate into a large data variability, compounded by uncontrollable factors during the acquisition process (such as atmospheric conditions), which must be properly managed for a fruitful analysis. Finally, processing methods must contend with the huge amount of RS data, which increases daily by more than tens of terabytes (TBs) per day (Saralioglu & Gungor, 2020; Xu, Du, Yan, & Fan, 2020), and a severe lack of labeled samples (annotation of RS data is time-consuming), which is insufficient to cover all the data variability (Li, Zhang, Li, Ye, et al., 2023). To tackle these challenges, multi-domain approaches have proposed efficient generalization methodologies (Wang, Xie, Cheng, Zhang, & Liu, 2023).

In this sense, and aiming to process large amounts of data, the scientific community has invested great efforts in combining ML/DL techniques with advanced Big Data analysis strategies and High-Performance Computing (HPC) (Moreno-Álvarez, Paoletti, Cavallaro, Rico, & Haut, 2022; Zhang, Zhou, & Luo, 2022). Such advancements have enhanced the processing performance for handling significant volumes and varieties of RS data (Han et al., 2023; S. K & Sivanandan, 2023).

1.2. The impact of machine learning in earth sciences

In the realm of Earth sciences, significant progress has been achieved in gathering the rich features present in the data. For this purpose, ML techniques ranging from supervised to unsupervised alternatives, serve as invaluable tools for interpreting and extracting such information with minimal human intervention (Haut et al., 2021), reaching outstanding results in land-cover applications (Kotsiantis, Zaharakis, Pintelas, et al., 2007; Singh, Thakur, & Sharma, 2016).

Conventional supervised techniques include statistically-based procedures, such as Maximum Likelihood Classification (MLC) and Multinomial Logistic Regression (MLR) (Haut & Paoletti, 2020; Hogland, Billor, & Anderson, 2013). The former assumes that the statistics of each class follow a normal distribution, minimizing the posterior probability of a sample belonging to a class, and the latter models the probability of a sample belonging to a class, using an optimizer to adjust the logistic coefficients. However, these methods suffer from some limitations, e.g., the assumption of a particular data distribution, the difficulty of handling high-dimensional data and large datasets, and the limited ability to find complex/non-linear relationships within the data. In contrast, data mining techniques extract meaningful insights from RS data by focusing on predictive modeling rather than descriptive analysis. In general, these are mostly assumption-free models, which are more robust to handling large dimensional data sets—although they may lack interpretability. For instance, decision trees (DTs) and random forests (RFs) have been widely used to conduct land-cover classification (Alcolea, Paoletti, Haut, Resano, & Plaza, 2020; Belgiu & Drăguț, 2016; Friedl & Brodley, 1997; Waske et al., 2012) due to their simplicity, flexibility, and computational efficiency, where each node defines a decision-making step that evaluates a specific condition on the data. These methods have been studied alongside other techniques, such as support vector machines (SVMs) (Adam, Mutanga, Odindi, & Abdel-Rahman, 2014; Sheykhoumoussa et al., 2020; Thanh Noi & Kappas, 2017). SVMs gained popularity in RS due to their efficiency in terms of accuracy, robustness to high dimensionality data, and indifference to the type of input data (Camps-Valls et al., 2004; Maulik & Chakraborty, 2017).

Most of the substantial advances in RS have come with the development of neural networks (NNs) models, experiencing significant progress in image analysis due to their ability to solve complex nonlinear problems (Atkinson & Tatnall, 1997). The origin of the initially so-called “shallow learning” lies in mimicking the information-processing

behavior of the human brain, wherein neurons communicate hierarchically through electrical signals (LeCun, Bengio, & Hinton, 2015). Among various NN architectures, the feed-forward back-propagation multi-layer perceptron (MLP) was the most widespread (Baraldi, Binaghi, Blonda, Brivio, & Rampini, 2001; Mas & Flores, 2008), serving as the foundation for current DL architectures (Ahmad et al., 2021; Paoletti, Haut, Plaza, & Plaza, 2019; Yuan et al., 2020; Zhong, Ling, & Wang, 2019). Currently, a wide variety of neural architectures have been implemented for different RS analysis, such as auto-encoders (AEs), to extract meaningful and compact representations from RS data (Khajehrayeni & Ghassemian, 2020; Ma, Wang, & Geng, 2016; Zhou, Shao, Diao, & Cheng, 2015); recurrent neural networks (RNNs), to handle sequential and temporal data (Maggiori, Charpiat, Tarabalka, & Alliez, 2017; Paoletti, Haut, Plaza and Plaza, 2020; Sharma, Liu, & Yang, 2018); convolutional neural networks (CNNs), to produce descriptive hierarchical features by combining spatial and spectral/temporal information (Ghanbari, Mahdianpari, Homayouni, & Mohammadimanesh, 2021; Paoletti, Haut, Plaza and Plaza, 2018; Sharma, Liu, Yang, & Shi, 2017); GANs, to generate, enhance, or simulate RS data (Liu, Zhou, Xu, Liu and Wang, 2020; Paoletti, Haut, Ghamisi et al., 2020; Roy, Haut, Paoletti, Dubey, & Plaza, 2021); capsule networks (CapsNets), to capture spatial relationships within data (Ma et al., 2019; Paoletti et al., 2018; Zhou, Zhou, & Liu, 2020); and transformers, to capture contextual relationships and long-range dependencies in data (Deng, Xu, & Huang, 2021; Roy, Deria, Hong et al., 2023; Roy, Deria, Shah et al., 2023; Xu, Zhang, Zhang, Yang, & Li, 2021).

This success of DL approaches is determined by multiple factors, e.g., their feature extraction ability, providing key data features without establishing prior assumptions about the data distribution. This reduces human efforts by avoiding the design of intricate hand-crafted features (Alzubaidi et al., 2021; Qiu & Jensen, 2004). Remarkably, CNNs have defined the current state-of-art in RS image processing due to their success when extracting discriminative features from raw data with relevant models developed over the years, such as the visual geometry group (VGG) (Ye et al., 2021), the residual network (ResNet) (Paoletti et al., 2018; Roy, Manna, Song, & Bruzzone, 2020; Zhu et al., 2021), or the dense network (DenseNet) (Liang, Zhang, & Li, 2022; Tong, Chen, Han, Li, & Wang, 2020), among many other implementations. Furthermore, based on the premise that deeper networks achieve better feature representation, the general trend in NN development has been to build deeper neural networks (DNNs) with millions and billions of parameters (Alzubaidi et al., 2021; Szegedy et al., 2015; Thapa, Horanont, Neupane, & Aryal, 2023). Indeed, CNN-based models have three main significant advantages: (i) the introduction of translation invariance, (ii) the reduction of the number of parameters to be tuned due to weight sharing, and (iii) the ability to learn spatial correlation. These have contributed to the near-standard use of CNN-inspired models to solve many RS challenges, such as scene classification (Adegun, Viriri, & Tapamo, 2023), object recognition (Sagar, Chen, Xie, & Kim, 2024) and segmentation (Li, Cai, Li, Kou and Zhang, 2024). Notwithstanding their excellent results, this should not disguise the fact that CNN-based models are extremely data-greedy and tend to degrade under limited labeled sample scenarios. Consequently, this data deficiency prevents models from encompassing the full range of data variability, thereby compromising the generalization ability to unobserved data (Dutta & Das, 2023). In this regard, collecting crowd-sourced data involves gathering information from multiple sources with high variability, while presenting particular challenges.

1.3. Crowd-sourced data: thoughts and insights

Extensive research has been conducted to overcome the lack of labeled RS samples (Paoletti, Mogollon, Moreno, Sancho, & Haut, 2023; Roy et al., 2021). In this sense, crowd-sourced data represents a valuable and natural solution, where volunteers — disjoint customers — collaborate in compiling extensive data to create sufficiently enriched

datasets. Thus, crowdsourcing offers rich, valuable data from different sources at a lower cost than traditionally annotated datasets, opening a new research field for image classification. Some of the most common crowd-sourced LULC datasets are presented in Table 1, with a noteworthy mention of the RSI-CB dataset (Li et al., 2020a). This dataset was specifically designed for classification purposes, providing global-scale RGB imagery.

The literature provides different methods for utilizing RS data from crowd-sourced datasets. In particular, multi-source data fusion techniques are gaining significant attention due to their ability to handle data with different properties (Li, Shi et al., 2022), wherein pansharpening is one of the hot topics (Deng et al., 2022; Du, Zheng, Lu, & Doudkin, 2021; Vivone et al., 2020). Also, the use of miscellaneous data extracted from social media has been recently studied (Ahmed, Huynh, Wickramasinghe, & Vu, 2015; Redondo et al., 2021; Shi, Qi, Liu, Niu, & Zhang, 2019), with DL-based fusion approaches offering a powerful tool for classifying and interpreting the content of the data (Han et al., 2022; Liu, Zheng et al., 2022; Xu et al., 2018). Despite the promising results, these methods face multiple barriers: high computational complexity, high data variability, and the generalization of unbalanced datasets. Indeed, a large number of learning parameters are arranged into deep and/or wide architectures to capture and represent data complexities effectively, involving large matrix calculations. Nonetheless, using crowd-sourced data addresses the issue of sample scarcity by offering access to an extensive array of information collected from unlimited sources. Furthermore, the complexity is exacerbated by the significant differences in crowd-sourced data, covering geographically distant areas and compounding both the intra-class variability (as they record different types of features that can expand multiple subcategories within the same class) and inter-class similarity (due to the existence of classes with similar properties in terms of color, shape, or texture). Specifically, crowd-sourced data generates different issues in the field of Earth observation, such as observation quality, inconsistency and disagreement with different land cover maps, legal and ethical issues, measurement protocols, and different data distributions (Huang, Li, Yang, & Zou, 2023). These issues lead to data discrepancies and imbalances for each land-cover class, thereby biasing the training of the aforementioned models. Recent developments in distributed computing and ML have mitigated these concerns for large-scale analysis (Wu, Tian, Zhang, Zeng, & Zeng, 2020). In future research, tackling these challenges correctly is expected to bring a substantial improvement in crowd-sourced RS data classification applications.

1.4. Federated learning as a promising solution

Distributed training has become the standard to handle data complexity motivated by the significant developments in HPC (Sharma, Sharma, & Jindal, 2021). Such platforms provide the computational and storage resources needed to efficiently accelerate DNNs by parallelizing and distributing computations. In this regard, multiple open-source distributed frameworks are available to exploit all the computational capabilities. End-to-end software like PyTorch (Li, Zhao et al., 2020) and TensorFlow (Abadi et al., 2015) are remarkable examples as they both support the most popular communication back-ends, e.g., Message Passing Interface (MPI), NCCL, and Gloo (Gloo, 2020; NVIDIA, 2017; Snir, 1998).

Several distributed strategies have been developed to fully exploit the computational resources, e.g., data and model parallelism. In the former, training data is divided among different processes, each equipped with a full replica of the model (Moreno-Álvarez et al., 2020); in the latter, the model is partitioned, while training data is propagated through each device (Moreno-Álvarez, Haut, Paoletti, & Rico-Gallego, 2021). Data parallelism is often used due to its simplicity and low communication overhead.

Table 1
Popular land-use and land-cover (LULC) crowd-sourced datasets.

Dataset	Data type	Samples	Scale	Source	Aim
RSI-CB (Li et al., 2020a)	RGB images	24,743	Global	Points of interest obtained from the OpenStreetMap; images downloaded from the Google Earth and Bing Map servers.	Scene and ground object classification.
Fritz et al. (2017)	Data records based on geographic locations	151,942	Global	Four campaigns using the Geo-Wiki crowdsourcing platform.	Reference data for global satellite-derived products.
Clark and Aide (2011)	Data records based on geographic locations	46,207	Latin America and the Caribbean	Google Earth within a web-based application; MODIS satellite data used.	LULC mapping algorithm calibration and accuracy assessment.
Bayas et al. (2017)	Data records based on geographic locations	35,866	Global	One campaign using the Geo-Wiki crowdsourcing tool; imagery from Google Maps, Microsoft Bing or OpenStreetMap; time series obtained from different satellite sensors.	Train classification algorithms for cropland cover.

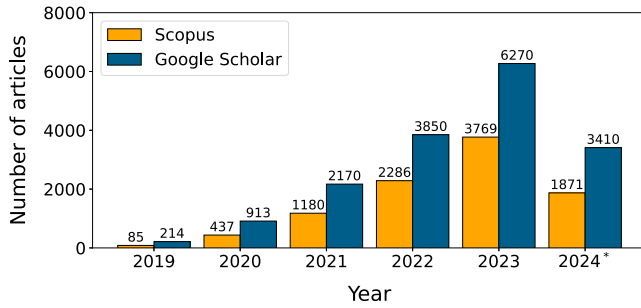


Fig. 1. Number of published research articles on FL per year according to Scholar and Scopus databases. A rapid increase in publications has been observed in the last six years. The data were collected on June 13, 2024, using the following keyword search: *intitle:“federated learning”* in Google Scholar, not including citations; *TITLE(“federated learning”)* in Scopus, not including preprints.

In such distributed strategies, two communication approaches play a key role: centralized and decentralized. In centralized schemes, a data center manages the coordination and exchange of information among all participants. This approach streamlines the communication structure but creates a bottleneck as the central server becomes the focal point for all data transfers. In addition, centralization compromises data privacy by transmitting sensitive information from multiple sources. In the RS context, this privacy is influenced by various factors, including the legal frameworks of different countries, the privacy and security mechanisms employed by companies, and the inherent sensitivity of the data. In summary, the centralized approach involves collecting, cleaning, and integrating data from different sources, which poses several challenges such as privacy preservation, data/system heterogeneity, and communication efficiency (Farkas, Kertész, & Lovas, 2020).

On the other hand, decentralized solutions are increasingly recognized for their efficacy in addressing inherent challenges associated with centralized systems, particularly in communication and data management. The concerns regarding privacy and security are mitigated in decentralized models, where data remains stored on local clients. This approach not only enhances privacy but also contributes to a more secure and robust system. Another benefit is that computing tasks can be distributed among nodes, reducing congestion and potential delays. The adaptability of decentralized systems is notable, allowing for more flexible implementation of changes without disrupting the entire network. Nonetheless, the choice between centralized and decentralized communication depends on the specific requirements of the application and the nature of the data, among others. Both approaches have proven to be efficient, resulting in notable savings in communications (Yuan, Wang, Sun, Yu, & Brinton, 2024).

In response to the growing disjoint nature of remotely sensed data, decentralized approaches emerged as an innovative solution. Concretely, FL addresses data privacy and security concerns, leveraging advancements in computational hardware to operate within an unlimited scalable environment. Moreover, FL meets the requirements and

regulations necessary to preserve data integrity while avoiding public exposure, thereby fostering the development of more efficient methods and potentially richer datasets. For instance, in military applications, these solutions facilitate collaboration among multiple military organizations while ensuring data privacy and security by not exchanging raw data (Arora, Khullar, Kansal, Kumar, & Popli, 2024). Moreover, the FL paradigm aligns naturally with real-time applications, as model training takes advantage of data from diverse geographical areas. This synergy offers significant scalability benefits, particularly in fast-response real-world scenarios.

All these aspects have contributed to FL gaining notable attention in recent years, as demonstrated in Fig. 1. This approach encompasses a set of ML techniques to collaboratively train models when the data are stored on different clients (e.g., remote devices or nodes). Furthermore, learning algorithms demand high-quality data, which is generally unavailable or spread across a potentially large number of devices, known as *data islands*. Interesting research has been conducted in RS (Aledhari, Razzak, Parizi, & Saeed, 2020; Li, Sahu, Talwalkar and Smith, 2020), where FL is used to support UAV-enabled wireless networks (Brik, Ksentini, & Bouaziz, 2020), in real-time scenarios (Tam, Math, Nam, & Kim, 2021), or in conjunction with Internet of Things (IoT) approaches (Fadlullah & Kato, 2021). Recent studies have demonstrated the efficiency and privacy-awareness of utilizing multiple devices in handling data discrepancies between clients (Li, Xie et al., 2024). Additionally, FL-based real-time applications have been deployed to efficiently process cooperative collected data in situ for satellite on-board ML (Han, Hosseinalipour, Love, Chiang, & Brinton, 2024; Razmi, Matthiesen, Dekorsy, & Popovski, 2024).

1.5. Summary of contributions

Given the characteristics of FL, it becomes imperative to focus on communication aspects, as they give rise to significant concerns regarding model accuracy and privacy. Moreover, secondary concerns stem from hardware specifications, which may impact communication and network status. Furthermore, the availability of devices may fluctuate over time. Additionally, the data distribution tends to be uneven among different devices.

The primary objective of this study is to assess the suitability, i.e., effectiveness, and performance of FL techniques in tackling the challenges associated with crowd-sourced RS data. Therefore, this research focuses on the innovative integration of these two domains, FL and RS. Nevertheless, it is also recognized the broader implications of addressing real-time data challenges, especially in contexts characterized by diverse data sources and dynamic environmental conditions. The contributions of this paper are as follows:

- A comprehensive review of novel FL strategies applied to remotely sensed imagery is provided. The research demonstrates the convergence of both fields through performance, privacy, and integrity studies in a scalable and fault-tolerant environment.

- A distributed decentralized training approach is introduced to address the challenges of Earth surface image data and the scarcity of labeled samples.
- The case study employs a crowd-sourced dataset processed by multiple clients across various communication topologies. A wide range of configurations are studied to demonstrate the generalization ability of the model for classification tasks.
- The study demonstrates that FL methodologies offer significant advantages in addressing data imbalances within heterogeneous environments.
- This paper aims to pioneer a new research direction for future contributions in the field of RS, tackling the inherent complexity of current model architectures, data growing and security concerns.

The remainder of this paper is structured as follows. Section 2 delves into the field of FL by introducing its formal definition, common architectures, and the main challenges addressed in this discipline. Section 3 bridges the gap between FL and RS to show the advantages of combining both domains. Section 4 presents the case study of this work, in which different FL techniques are used to evaluate their effectiveness when classifying RS data. Finally, Section 5 concludes this work and offers diverse observations.

2. Federated learning concepts: An overview

The distributed nature of FL not only encourages the exploration of solutions to constraints and challenges in collaborative ML model training but also establishes a secure business framework for fostering the development of ML/DL-based solutions among companies (Li, Wen et al., 2021). FL facilitates the creation of robust models based on processing distributed and potentially heterogeneous datasets while preserving data privacy (Yang, Liu, Chen, & Tong, 2019). This paradigm has been implemented across various tasks, including yield forecasting (Li, Markovic, Edwards and Leontidis, 2024), classification (Zhu, Wu, Bashir, Pan, & Yang, 2023), and semantic segmentation (Dong et al., 2023; Miao, Yang, Fan, & Yang, 2023).

Generally, FL is categorized into two primary types: model-centric and data-centric approaches. The former, which is notably more common (Li, Fan, Tse and Lin, 2020), is the central focus of this overview.

2.1. Main components involved

The concept underlying model-centric FL involves decentralized and collaborative training of an ML model parameters θ , using disjoint datasets $\{D_1, \dots, D_N\}$ distributed among N clients. The aim is to achieve a final accuracy comparable to that attained through traditional centralized training on a merged dataset $\{D_1 \cup \dots \cup D_N\}$. Nevertheless, this scenario poses complex challenges, including constraints such as privacy preservation or data ownership. In general, collaborative training depends on three key components: clients, the manager, and the communication design.

Clients, also known as data owners or participants, manage the data and train local models, ensuring privacy by not directly exposing private data and, ultimately, benefiting from the final global model. The nature, size, and amount of clients define the scaling nature, which is classified into two main categories: (i) *cross-silo*, where the clients are formed by a relatively small number of companies, organizations, or data centers with powerful computational resources and considerable amounts of data available; and (ii) *cross-device*, in which a potentially large number of devices participate in the training process with limited data and computational power.

The *manager*, also referred to as the server, is tasked with collecting intermediate results from the clients. The responsibility of the manager lies in holding the aggregated model based on the collected data. In cross-device scenarios, a computationally powerful central server

typically assumes the role of the manager. However, in cross-silo, one of the clients normally undertakes the tasks related to the manager.

The decentralized communication design omits the global server (Kairouz et al., 2021a; Li, Sahu et al., 2020) and communications are established solely between neighboring clients. As mentioned, this configuration addresses the bottleneck issues associated with star-shaped networks. In this context, various communication approaches based on decentralized data have been developed to address the complex dynamics inherent in distributed systems. As an example, in hierarchical structures, nodes are organized into tiers, each with specific roles and responsibilities. This organized framework enables efficient information flow, ensuring that communication occurs seamlessly across different levels.

2.2. Data partition scheme

Let $D_i = (S_i, F_i, L_i)$ be the dataset stored on the i th client among N clients with S , F , and L representing the sample, feature, and label data spaces, respectively. In this context, D_i can be visualized as a matrix where the horizontal axis contains the samples, while the vertical axis indicates distinct features corresponding to each sample within the dataset. Given the different possibilities for data distribution among clients, three categories of FL have emerged, as shown in Fig. 2: horizontal FL, vertical FL, and federated transfer learning.

Horizontal FL, also known as sample-based FL or homogeneous FL, represents a natural approach to data partitioning in which data are distributed among clients along the sample dimension. This implies that the datasets stored on the clients differ in the sample space but share the same feature and label spaces ($S_i \neq S_j$, $F_i = F_j$, $L_i = L_j$, $\forall i, j \in [1, N]$, $i \neq j$). Consequently, overlap in data samples is minimal, and clients maintain identical data representation (features and labels). With this approach, the number of available training samples increases considerably, potentially leading to improved performance of the overall model.

Vertical FL, also referred to as feature-based FL or heterogeneous FL, partitions data among clients by the feature dimension. In simpler terms, each dataset shares a common sample space but notably diverges in both the feature and label spaces ($S_i = S_j$, $F_i \neq F_j$, $L_i \neq L_j$, $\forall i, j \in [1, N]$, $i \neq j$). This approach ensures that the training set maintains the same number of samples while expanding the feature dimension by incorporating non-matching columns. Consequently, it facilitates a broader and more diverse data representation across participating clients. This diversity can significantly enhance model generalization and learning capabilities, particularly in scenarios where it is crucial to combine features while preserving data privacy and integrity.

In contrast to the previous scenarios, where data typically share either the sample space or the feature or label space, the domain of federated transfer learning (Liu, Kang, Xing, Chen, & Yang, 2020) presents an alternative setting. Here, the datasets stored on the clients exhibit substantial disparities across all spaces: sample, feature, and label ($S_i \neq S_j$, $F_i \neq F_j$, $L_i \neq L_j$, $\forall i, j \in [1, N]$, $i \neq j$). This disparity presents two different concerns: the absence of data labels and poor data quality. For this purpose, federated transfer learning involves transferring the knowledge acquired from the source domain (derived from processing non-overlapping data) to the target domain, which is commonly related, thereby enhancing performance. This approach emerges as a powerful strategy to overcome the limitations posed by data islands while strengthening data security and user privacy.

2.3. Model performance and data integrity

The performance of FL has been demonstrated in diverse tasks, showcasing scalable and effective methodologies on decentralized data sources (Xiong et al., 2024). Furthermore, enhancing the generalization performance across heterogeneous domains and achieving robust learning at local client level remain persistent challenges in building

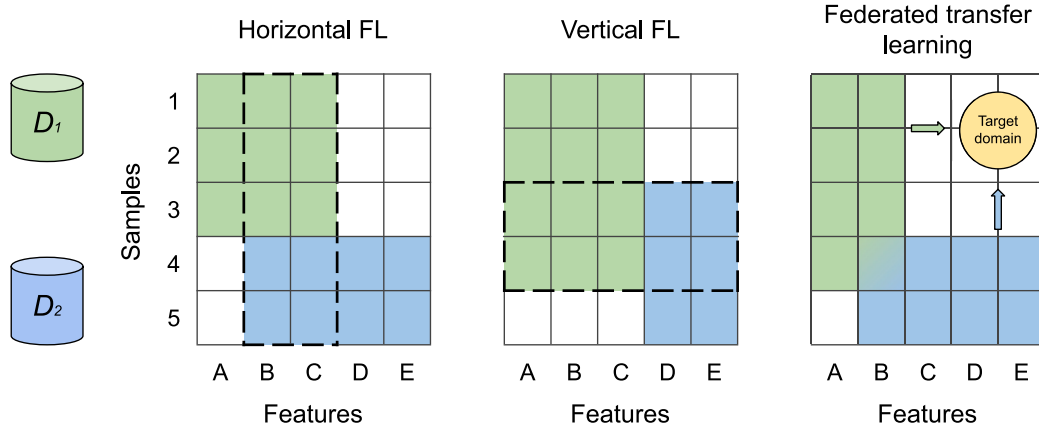


Fig. 2. Classification of federated learning (FL) according to data distribution. Each cell represents a specific value of a characteristic (column) in a sample (row) within its corresponding dataset (D). Cells grouped using dashed lines correspond to the training data. For clarity, a total of five samples and features are displayed.

accurate models (Xie, Dai, Wang, Li, & Zhao, 2024). In RS applications, this generalization ability has been deeply studied for heterogeneous datasets to minimize the category-wise feature distributions across different client models (Zhang, Zhang, Yu, & Kang, 2023) and maximize image similarity within categories (Buyuktas, Sumbul, & Demir, 2023). However, the characteristics of crowd-sourced RS datasets have not been explored in the literature within the context of FL. This study represents an initial step in addressing this gap, recognizing the similarities between the two fields.

In this regard, performance and integrity in such complex environments are compromised by the presence of malicious participants. These participants may adversely affect the performance of the global model and pose significant risks to data privacy. These threats are classified according to their category or type of attack, with corresponding defense mechanisms being crucial for mitigation. Such measures are essential to safeguard against potential disruptions and ensure the robustness of FL models, particularly when applied to sensitive and heterogeneous datasets like those in RS. Concretely, potential threats within FL environments generally fall into two categories (Lyu et al., 2022; Yang, Fan, & Yu, 2020): poisoning and inference attacks.

In *poisoning* attacks, the aim is to corrupt the aggregated ML model during the training stage (Bagdasaryan, Veit, Hua, Estrin, & Shmatikov, 2020). This is accomplished through (i) *data poisoning*, where an adversarial client injects malicious data into the training process to distort the learning process; and (ii) *model poisoning*, where the attacker manipulates the training process by altering the parameters or gradients of the local model to compromise its performance. Additionally, poisoning attacks can also be grouped into two categories depending on the target of the attack. First, *targeted attacks*, such as backdoor attacks, aim to corrupt the aggregate model with specific alterations deployed by a malicious client that could trigger certain behaviors (e.g., bias toward classification of a specific class) under particular conditions without compromising overall performance. Second, *untargeted attacks*, also known as Byzantine attacks, seek to hinder the convergence of the aggregated model, thus affecting performance.

On the other hand, *inference attacks* capitalize on the potential indirect exposure of sensitive information during the communication process (Aledhari et al., 2020). This occurs despite the direct prevention of data leakage inherent in FL approaches, where models are trained locally, and participants share only model updates — gradients or weights — with the manager rather than the original raw data. In practice, inference attacks can actively or passively extract information from the trained model (Nasr, Shokri, & Houmansadr, 2019). These attacks are categorized as either *white-box*, where the adversary has complete access to the target global model's architecture and parameters, or *black-box*, which rely on limited information about the model, such as its input-output behavior.

Multiple defense mechanisms have been proposed to safeguard the learning process in an FL environment (Lyu et al., 2022; Zhang et al., 2021). These mechanisms aim to ensure privacy without significantly increasing the complexity of the aggregated model or compromising its accuracy. Some of the most popular methods proposed in this field include encryption techniques like *homomorphic encryption* (Acar, Aksu, Uluagac, & Conti, 2018) and approaches based on adding random noise to sensitive information such as *differential privacy* (Brendan McMahan, Ramage, Talwar, & Zhang, 2017). Concretely, homomorphic techniques are based on encrypting model updates before transmission. This process ensures confidentiality by allowing computations using encrypted data, producing results that, upon decryption, match those derived from operations on the original data. In practical terms, an encryption scheme is termed homomorphic over an operation $\star$ if $E(m_0) \star E(m_1) = E(m_0 \star m_1)$, $\forall m_0, m_1 \in \mathcal{M}$, where $\mathcal{M}$ denotes the set of possible messages and E the encryption algorithm. Moreover, differential privacy involves the addition of random noise to the local model gradients generated based on a specified data distribution. Similar to the previous method, such technique aims to prevent the reconstruction of individual data or participant contributions while preserving the integrity of the outcomes of the model.

In addition to the aforementioned, FL must tackle further challenges derived from its heterogeneous nature, such as addressing communication issues, handling client dropouts, and managing data heterogeneity.

2.4. Heterogeneous federated learning

The training of FL models relies on substantial amounts of data distributed among a potentially expanding number of clients. This fact is accentuated for crowd-sourced data, where clients typically possess significantly diverse heterogeneous characteristics. Also, such clients may unexpectedly join or leave the training process, impairing the training performance and, consequently, the convergence of the model. Following this, a description of the heterogeneous challenges identified in the training process and their impact on performance is presented.

Firstly, the gradient communication between clients may be compromised by the presence of heterogeneous devices during training. In particular, straggler clients, characterized by slower or less reliable communication or computational capabilities, can significantly degrade the overall performance of the global model (Moreno-Álvarez et al., 2022). Devices with slower computing are more prone to fail to complete the training within the required time. This can result in longer waiting times for aggregation or even non-reporting results, ultimately leading to decreased performance. To deal with this drawback, asynchronous schemes have been proposed to mitigate the negative impact of stragglers (Xu, Qu, Xiang, & Gao, 2023). Moreover, fault tolerance strategies, inspired by classical distributed HPC systems, have been

adapted and implemented in FL environments (Wang et al., 2019). For example, ignoring client failures is an empirically simple solution that could introduce biases in training. Alternatively, sophisticated fault tolerance mechanisms have been proposed, including active sampling to handle heterogeneity by selecting those clients who participate in each training round (Li et al., 2018).

Secondly and linked to the aforementioned challenges, the major obstacle is the management of the diverse device capabilities, i.e., limitations of computing power, communication, connectivity, storage capabilities, and power consumption, among others. For instance, communication reduction techniques (Chen, Sun, & Jin, 2019; M Abdelmoniem, Elzanaty, Alouini, & Canini, 2021) avoid the bottlenecks caused by the waiting times of faster processes at communication points. Other techniques to handle communications and vast data volumes involve compressing model updates to reduce transmitted message sizes, selectively dropping devices to minimize communication, and implementing decentralized training.

Additionally, the wide range of heterogeneous devices capturing data leads to non-independent and identically distributed (non-IID) data, i.e., the data stored on each client are not representative of the overall collection. This problem, which arises due to differences in resource availability among devices, causes the manager to receive dissimilar model update information. Although the aggregated model is expected to accommodate these disparities effectively and generalize accurately, this divergence adversely impacts both model convergence and accuracy (Tijani, Ma, Zhang, Jiang, & Doss, 2021). The work in Kairouz et al. (2021b) thoroughly classifies and describes different dimensions of this problem. For instance, clients with larger volumes of data contribute more significantly to the aggregated model, thereby introducing bias toward their data. Also, the studies (Li, Diao, Chen and He, 2022; Li, Huang, Yang, Wang, & Zhang, 2019) investigated these scenarios, concluding that the system heterogeneity affects the convergence of the FL models. Therefore, the nature of data skew significantly influences the effectiveness of these models (e.g., label data skew is more influential than quantity imbalance).

Finally, several studies have suggested different solutions for managing non-IID data. The work Tijani et al. (2021) introduces a data-sharing strategy for the clients to share a small portion of their private data with the other participants. Alternative methods include incorporating correction terms into model updates or maintaining multiple global models tailored to distinct data distribution groups to address data skew. In this regard, the work Verma et al. (2019) discusses how data skew affects FL problems, evaluating further solutions based on aggregating related features to model updates (bounds-aware fusion) and increasing features by interchanging representative data among clients. As a relevant example, FedProx (Li et al., 2018) presents a framework that offers theoretical guarantees for learning over non-IID data, with each participant performing a variable amount of work. FedProx achieves convergence comparable to FedAvg — the widely used framework in the FL domain — in realistic FL datasets, demonstrating that the generalization capability is determined by such levels of heterogeneity. Specifically, the performance of the global model, denoted by parameters Θ , is mainly influenced by the collaboration of clients in each training iteration, as depicted in the computation from Eq. (1). The objective is to minimize the loss ℓ across the available sample space data S_i contributed by each client i during the training communication round t , considering the client aggregation weight p_i (proportional to the size of training data). Note that only $N_t < N$ clients participate in a specific round.

$$F_{collab}(\Theta) := \frac{1}{N_t} \sum_{i \in N_t} [p_i \cdot (\frac{1}{|S_i|} \sum_{s \in S_i} f(\Theta; o))] \quad (1)$$

Overall, this data heterogeneity inherent in FL architectures finds an ideal partner in the domain of RS applications. This synergy establishes the groundwork for a harmonious collaboration aligned with the requirements and challenges presented by the data-intensive nature of RS scenarios. The following section delves into the convergence between FL and RS, offering a detailed description that highlights the growing interest in this innovative topic.

3. Federated learning and remote sensing: Synergies and convergence

Exploring the convergence of FL and RS domains reveals valuable insights into enhancing real-world applicability. This analysis places special emphasis on the significant challenges posed by this union.

As previously mentioned, a key advantage of FL lies in ensuring data privacy for each client. In this regard, RS data is subject to multiple factors, encompassing international and national regulations. The regulatory landscape is additionally shaped by oversight from organizational entities tasked with licensing and imposing restrictions on satellite and airborne devices. Image acquisition involves capturing missions conducted by organizations with distinct objectives. This also poses significant challenges in processing, safeguarding, and distributing data among diverse clients. Thus, such organizations establish their own policies and regulations for using and handling remote sensors, primarily to address potential security threats.

Substantial benefits arise from effectively addressing privacy concerns, especially considering the vastness and diversity inherent in such decentralized scenarios. Currently, dozens of countries and hundreds of organizations leverage the rich information compressed within the data scenes. Therefore, efforts are being made to take full advantage of all available data, rather than avoiding the potential risks associated with sharing sensitive information. In such scenarios, it is imperative to ensure that no client-specific information is disclosed and that the outcomes are not unduly influenced by any such information. This could significantly enhance various application domains, including natural disaster detection, crop management, migration analysis, and military applications. Concretely, for image classification applications, the influx of substantial data inputs to models facilitates robust detection and extraction of image features. This translates into a marked improvement in the classification accuracy of unseen data.

Alongside privacy concerns, processing such data requires computing devices capable of efficiently handling vast volumes of information. The extraction of information relies on deploying complex models, entailing substantial computational and memory resources. As aforementioned, this is primarily attributed to the expansive dimensions of RS data, the sheer volume of data, and the costly processing operations. As a solution, FL offers a computing environment capable of mitigating the high costs associated with communications and computations. Also, through distributed processing, FL establishes an efficient environment to avoid excessive transmission and non-optimal communication schemes. This marks a departure from data communication techniques employed in cloud or HPC platforms, where continuous message flow can lead to communication channel overload and a lack of privacy, as established techniques for privacy control or data location management are often absent. Computational workload is determined by Eq. (2), being BS the batch size and I_i the number of iterations of client i .

$$S_i = \sum_{n=0}^{I_i} BS, \quad I_i = \frac{S_i}{BS} \quad (2)$$

As a result, data-separable computation can be conceptualized as a collection of independent subtasks, each corresponding to different segments of the overall dataset. In essence, the computation performed by each client depends solely on its own data. This approach can be represented by each client computing $f_i(S_i) \in F_{collab}$, based on the specific algorithm (Zhou & Li, 2023). Consequently, the computational complexity can be expressed as $O((\beta + n)N \cdot I)$, where β denotes the fixed overhead associated with the gradient calculation of the specific algorithm, and n represents the variable cost related to the size of the client sample data S_i .

Furthermore, addressing mismatches in the total training data amounts is crucial to building robust models that handle different captured areas. Some of these diverse datasets are detailed in Table 2. For instance, multi-source datasets like AID, RESISC45, and WHU-RS19 face challenges related to predicting common and mismatching classes.

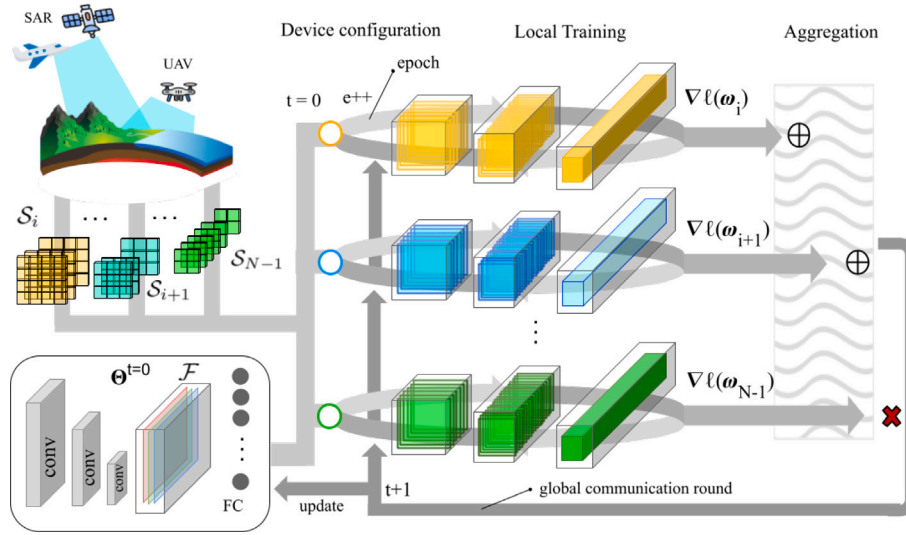


Fig. 3. Federated learning process using heterogeneous clients $i \in N$ and data S_i . The objective is to train an initial model parameters $\theta^{t=0}$ by aggregating local models gradients $\nabla \ell(\omega_i)$ to build a global model θ^{t+1} . Data are captured using different devices. The client configuration takes place once. Each client trains private data locally during a selected number of epochs e . The local training time depends on heterogeneous processing capabilities. Unselected clients (stragglers) are discarded from the aggregation stage, dropping their information.

Table 2

Remote sensing scenes that collect imagery from multiple sources for land-use classification. C and R denote the images per class and spatial resolution, respectively. The indicated sources are Google Earth (GE) and crowd-sourced (CS). Overall accuracy (OA) and cross-domain overall accuracy (CD-OA) values are obtained from Gominski, Gouet-Brunet, and Chen (2022).

Attribute	AID	RESISC45	RSI-CB	WHU-RS19
Images	10,000	31,500	24,743	1005
Classes	30	45	35	19
C	200–400	700	200–1300	50
Size	600×600	256×256	256×256	600×600
R (m)	0.5–0.8	0.2–30	0.3–3	<0.5
Sources	GE	GE	CS	GE
Year	2016	2016	2020	2012
CD-OA (%)	81.57	71.04	88.16	97.05
OA (%)	97.21	95.17	99.66	97.50

As a solution, FL methodologies allow for a tolerant environment and the utilization of training data with non-matching representation features or data size. This approach has been previously demonstrated using transfer learning with ImageNet (Deng et al., 2009). Such observations suggest that classification algorithms can notably enhance their performance when fed with a sufficient volume and variety of images.

The intersection of these fields becomes notable as FL endeavors to tackle the aforementioned concerns associated with distributed data training. This serves as a cornerstone in the creation of innovative methodologies. The comprehensive procedure is visually represented in Fig. 3, which highlights how FL effectively manages the intricacies of the entire distributed process.

4. Case study: Image classification benchmarking

This case study presents and evaluates the performance of multiple FL methodologies in classifying optical land-use images across various domains using a crowd-sourced RS dataset. This section explores the convergence between both fields.

4.1. Dataset description

The RSI-CB (Li et al., 2020b) crowd-sourced dataset is built from high-resolution ($256 \times 256 \times 3$) and ($128 \times 128 \times 3$) RS images, generated using raw data types such as latitude, longitude, attributes,

and information from Open Street Map. The construction of the dataset involves multiple steps: (i) *data capturing*, where the aforementioned raw information and image data are jointly obtained; (ii) *filtering*, applied to the complete captured information while ensuring the relation between images and labels; (iii) *cropping*, enforcing fixed-size of pixel blocks by using raw image information; and (iv) *cleaning*, where a manual purge is performed for image blocks containing multiple categories. In this regard, the predominant information is selected, and inconsistent raw data is removed.

RSI-CB256 gathers data corresponding to multiple parts of the world with different categories and characteristics. Several cities with high data distributions have been selected to create this dataset: U.S. cities, such as New York and Washington D.C; Asian cities, like Beijing or Shanghai; or European cities, such as London or Liverpool, among others. However, crowd-sourced datasets face some challenges when performing classification, including density distribution, increased redundancy among data, and imbalance between categories predominant in a specific geographic area. Indeed, for RSI-CB, the U.S., China, Russia, Spain, and the U.K. dominate places for extracting raw interest data.

The RSI-CB dataset contains approximately 24,000 images divided into 35 categories captured over a range of 0.22–3 m per pixel. These categories represent different class labels, some belonging to a main category. As an example, the main category *Water sea* contains seven subcategories: *coastline*, *dam*, *hirst*, *lakeshore*, *river*, *sea*, and *stream*. The dataset is described in Table 3.

4.2. Experimentation: interface and methods

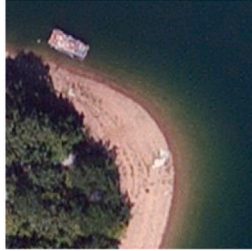
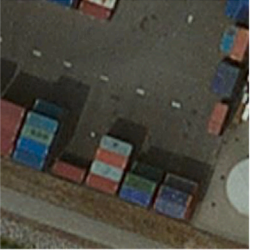
The experimentation was conducted using the Foundational Ecosystem Design for ML (FedML) API¹ (He et al., 2020) on the RSI-CB256 dataset. FedML allows the simulation of large-scale FL environments for distributed training. Thereby, several clients are launched to process a set of disjoint data following the behavior and properties of the FL process.

The FedML library follows the structure illustrated in Fig. 4. At its core components (Level 0), there are two primary packages. The first package comprises the communication module, which implements

¹ Available at <https://fedml.ai/>.

Table 3

RSI-CB256 dataset description for the main categories (left) and subcategories (middle-right). A total of 35 subcategories are detailed, along with their respective number of samples. The top of the figure shows four random samples.

								
(a) 3-Lakeshore	(b) 16-Airplane	(c) 22-Avenue	(d) 33-Container					
Category	Samples	Subcategories	Subcategory	Samples	Subcategory	Samples	Subcategory	Samples
0-Constructions	3791	4	0-Stream	688	12-Forest	1082	24-Snow mountain	1153
1-Cultivates	2817	3	1-Sea	1028	13-Artificial grass	283	25-Sandbeach	536
2-Other land	3593	4	2-River	539	14-Town	335	26-Mountain	812
3-Transportation	3300	7	3-Lakeshore	438	15-Pipeline	198	27-Desert	1092
4-Other objects	884	3	4-Hirst	628	16-Airplane	351	28-Green farmland	644
5-Woodland	6258	7	5-Dam	324	17-Parkinglot	467	29-Dry farm	1309
6-Water	4104	7	6-Coastline	459	18-Marina	366	30-Bare land	864
			7-Sparse forest	1110	19-Highway	223	31-Storage room	1307
			8-Shrubwood	1331	20-Crossroads	553	32-Residents	810
			9-Sapling	879	21-Bridge	469	33-Container	660
			10-River forest	524	22-Avenue	544	34-City building	1014
			11-Mangrove	1049	23-Airport runway	678		

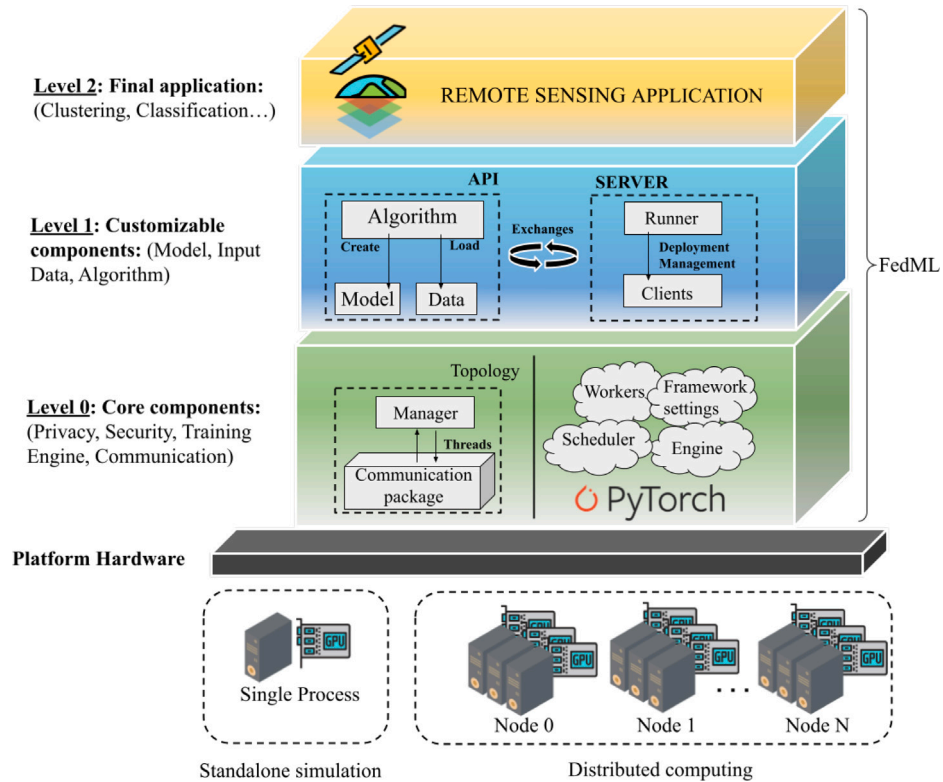


Fig. 4. Overview of the Foundational Ecosystem Design for ML (FedML) library architecture components for remote sensing applications (He et al., 2020). The hardware level depends on the established computing platform.

low-level communication mechanisms based on MPI, NCCL, or RPC, as well as security and privacy features like Attack-Defend approaches. A manager coordinates the exchange of messages between clients. FedML supports multiple system-wise configurations in addition to classical topologies, including centralized, decentralized, ring, and hierarchical approaches. The second package is the engine, which facilitates the configuration of training parameters like optimizers and loss functions. The next level (Level 1) stores data, models, and algorithms, allowing customization by the user. The server is also deployed at this level to collect data from clients. Finally, the application paradigm is established at the highest level (Level 2).

FedML demonstrates versatility across multiple platforms, showcasing a significant degree of adaptability. Specifically, this case study encompasses two distinct experiment configurations (refer to the bottom of Fig. 4). The objective is to leverage the FedML framework, transitioning from controlled simulations to real-world RS applications. To this aim, two different methodologies are under study, each featuring distinct aggregation algorithms outlined in Table 4.

The *Standalone Simulation*, denoted as *SSFed*, aims to deploy a local single-process setup for individual clients on a designated computational device, enabling an examination of the initial classification performance. In this context, clients are launched on the same device to provide preliminary insights. This simulation environment is deliberately designed to emulate the decentralized nature inherent in FL approaches. Nevertheless, data partitioning ensures that each simulated client possesses a distinct subset of the dataset. This setup mimics the data distribution among decentralized edge devices in a real-world scenario. Concretely, a centralized aggregator is emulated to handle the collection and integration of model updates. Multiple aggregation algorithms, including *SSFed_{DEC}*, *SSFed_{AVG}*, *SSFed_{HIE}*, *SSFed_{VER}*, and *SSFed_{TUR}* are considered for this purpose. These algorithms facilitate the aggregation of model updates across simulated clients. In this regard, the evaluation encompasses both individual and collective impacts on the learning process. Insights derived from this simulation lay the groundwork for scaling experiments to a more realistic distributed environment. Additionally, these simulations facilitate the exploration of different federated methodologies before their implementation in real-world applications.

On the other hand, *Distributed Computing*, referenced as *MPIFed*, provides the configuration to deploy a realistic workload parallelization and data distribution between processes. The platform consists of multiple nodes, comprising a total of N computational devices, referred to as clients. This configuration leverages parallel processing by deploying a specified number of processes, each on a different device. The experimental setup reflects the complexities of distributed systems with multiple real clients, offering insights into the performance and scalability of multiple federated methodologies. This package is composed of three approaches that includes *MPIFed_{AVG}*, *MPIFed_{OPT}*, and *MPIFed_{PROX}*. The communication approach is facilitated through MPI, which provides the backbone for the interaction of the processes. By deploying and assessing three distinct methodologies, this experiment contributes to understanding the challenges and opportunities posed by distributed computing.

Both alternatives for the corresponding algorithms are comprehensively delineated below, with descriptions including a thorough examination of the main features inherent in each alternative.

4.2.1. Preliminary assessment: Standalone simulations

The differentiation between decentralized and centralized approaches primarily lies in the method by which model updates are aggregated. As a summary of the previously introduced, in a centralized setup, all individual client updates are transmitted to a central server. Conversely, decentralized data averaging involves computing the average of model updates locally on individual client devices before transmitting the information. The studied methods include a novel decentralized approach, denoted as *SSFed_{DEC}* (Liu, Chen et al.,

Algorithm 1 Example of a FedML client-oriented communication approach following a centralized scheme. Communication rounds are performed in parallel by the clients.

```

1: Requires: Client  $i$ , communication rounds  $T$ , global server parameters  $\Theta$ , local weights  $\omega$  & Trainer trainer
2:
3: for  $t \in [1, T]$  do                                     ▷ communication rounds
4:    $\omega_i^t \leftarrow \text{get\_server\_model}(\Theta^t)$                  ▷ get model from server
5:    $\omega_i^{t+1}, I_i \leftarrow \text{trainer.train}(\omega_i^t)$ 
6:    $i \leftarrow \text{get\_client\_id}()$ 
7:    $\text{server} \leftarrow \text{get\_server\_id}()$ 
8:    $m = \text{Message}(\_, i, \text{server})$ 
9:    $m.add\_data(\omega_i^{t+1}, I_i)$ 
10:   $\text{send\_message}(m)$                                      ▷ send model to server
11: end for

```

2022). This specific decentralized methodology implements inter-node communications of the respective local training updates. In this scenario, striking a balance between communication efficiency and model convergence is crucial to minimize training complexity.

Traditional federated approaches involve one local update and single inter-node communication in each round. With the expansion of datasets and the increase in the number of clients, communication efficiency decreases. To address this limitation, the *SSFed_{DEC}* approach conducts multiple local updates on each client before globally sharing the gradients. This strategy enhances communication efficiency, promotes more robust gradients, and contributes to the identification of global optimal parameters. The performance of this approach has been demonstrated in different classification methods and applications (Moreno-Álvarez, Paoletti, Cavallaro, & Haut, 2023). Consequently, the methodology incorporates the discernment of discrepancies in the averages of models between neighboring clients. The local parameter update step and the respective communication are performed for each round, as shown in Eq. (3), where ω_i denotes the client i weights and $c_{(i \leftarrow j)}$ the confusion matrix of j clients involved in the neighbor-to-neighbor communication with client i . The confusion matrix represents the network topology for the consideration of each client's contribution to the global parameters. Then, the global parameter $\Theta^t = \sum_{i=0}^{N-1} \omega_i^t$ update applies the stochastic gradient descent (SGD) technique, calculated in Eq. (4), with C representing the communication confusion matrix, ∇ the gradients, t the corresponding time step, and α the learning rate. Communication compression has shown promise in enhancing performance within decentralized environments. However, its applicability in RS applications is still being studied, considering factors such as data complexity and the influence of unseen scenes by clients.

SSFed_{AVG} (McMahan et al., 2017) conducts a communication approach through a centralized server. The server collects data, distributes parameters between clients, and handles massive communications. Clients conduct local gradient descent using private data during the established learning iterations. Meanwhile, the server averages the resulting local model parameters from all clients on a weighted basis based on the number of data samples I to obtain the global parameters Θ . Eqs. (5) and (6) resume this behavior from the client and server perspectives, respectively, where $\mathcal{L}$ denotes the loss function. Algorithm 1 implements this approach. Notably, certain clients may execute a differing number of local updates, contingent upon their respective computational speed and data workload.

The high communication overhead poses a challenge in managing data transfers between clients. In this regard, *SSFed_{HIE}* (Wang & Wang, 2022) sets up an approach to handle stragglers clients and reduce the number of communications. Clients are arranged in a hierarchical network and established to send individual results asynchronously for the global parameter calculation. Each client i has a group $q \in Q$ of sub-clients deployed to perform training on local data.

Table 4
Survey of studied algorithms in the experimental research.

Algorithm	Equation	Abbreviation
Decentralized (Liu, Chen and Zhang, 2022)	$\omega_i^t = \sum_{j=0}^{N-1} c_{(i \leftarrow j)} \omega_j^t \quad (3)$ $\Theta^{t+1} = \Theta^t C^t - \alpha \nabla^t \quad (4)$	DEC
Averaging (McMahan, Moore, Ramage, Hampson, & y Arcas, 2017)	$\omega_i^{t+1} = \omega_i^t - \alpha \nabla \mathcal{L}(\omega_i^t, b_i^t) \quad (5)$ $\Theta^{t+1} = \sum_{i=0}^{N-1} \frac{I_i}{I} \omega_i^{t+1} \quad (6)$	AVG
Hierarchical (Wang & Wang, 2022)	$\omega^{t+1} = (1 - \alpha^t \omega)^t + \alpha^t \omega_q^{t-1} \quad (7)$	HIE
Vertical learning (Chen, Du, Lu, Wu, & Hung, 2022)	$\min_{\Theta} \mathcal{L}(\Theta; S) = \frac{1}{I} \sum_{k=0}^{I-1} \mathcal{L}(H_{s_k}, L_k^N) + \lambda \sum_{m=0}^{M-1} \gamma(\omega_m) \quad (8)$	VER
Turbo (So, Guler, & Avestimehr, 2020)	$\Theta^{t+1} = \sum_{g \in \mathcal{G}} \sum_{v \in \mathcal{V}} \omega_{(g,v)}^t \quad (9)$	TUR
Optimized (Asad, Moustafa, & Ito, 2020)	$\delta(\hat{\nabla}_{agg})^t = \sum_{i=0}^{N-1} \nabla \mathcal{L}(\omega_i^t, b) \quad (10)$	OPT
Proximal (Li et al., 2018)	$\min_{\omega} f_i(\omega^t; \omega^{t+1}) = SGD_i(\omega_i^t) + \frac{\kappa}{2} \ \omega_i^t - \omega_i^{t+1}\ ^2 \quad (11)$	PROX

Messages stream from the sub-clients to the client for the corresponding gradient aggregation operations. These calculations are performed in the server as shown in Eq. (7). The new model parameters are sent to the children clients. In this methodology, the learning rate α changes according to the central server updates.

SSFed_{VER} (Chen et al., 2022) implements a vertical federated taxonomy. In this setting, disjoint parties $m \in M$ composed of multiple clients maintain ownership of local data, ensuring the privacy and confidentiality of data and model parameters. The minimization objective in vertical learning is described in Eq. (8), where similar to previous approaches, each party m calculates local parameters and sends such information to the server. In this regard, $k \in I - 1$ corresponds to the different samples, the classification loss function is denoted as $\mathcal{L}(\cdot)$, H_{s_k} denotes the feature embedding and L_k^N denotes the respective labels. Note that feature information is decomposed over N clients.

SSFed_{TUR} (So et al., 2020) addresses the inherent scaling problem in FL. In this scenario, the massive increase in the number of clients causes an overload in the data aggregation step. This approach achieves safer aggregation by tolerating heavy client drops and reduces latency. For this purpose, a circular communication scheme divided into $\mathcal{G}$ groups is proposed. Privacy is ensured using private client random masks only known by the particular client and server. This method adds redundancy to the aggregation process by using Lagrange coding. In this method, information navigates from one group to another before culminating in the final model, where it is decoded. The parameter aggregation procedure is shown in Eq. (9) for the set of clients $\mathcal{V}$ remaining undropped.

4.2.2. Streamlining distributed computing

As previously introduced, distributed computing within HPC environments serves as a promising approach for FL-based applications. The search for efficient methodologies aims to implement strategies that enhance scalability and performance. As an initial approach, *MPIFed_{AVG}* implements the distributed version of *SSFed_{AVG}* where multiple processes work in parallel to take advantage of a distributed environment. The server is responsible for managing the processes on the platform and ensuring privacy. The global network parameters Θ^{t+1} are calculated by averaging information, as previously shown in Eq. (6).

MPIFed_{OPT} (Asad et al., 2020) presents an optimized algorithm to improve communication efficiency while preserving security and privacy. A sparse gradient top- k compression and encryption is implemented based on the proportion of each client's contribution to the gradient. The decryption δ of the global encrypted gradients $\hat{\nabla}_{agg}$ received from the server is performed through Eq. (10) in each client. Afterwards, the ω_i^{t+1} parameters are updated and aggregated to the server following Eq. (5).

Algorithm 2 FedProx proposed scheme.

- 1: **Requires:** Clients N , communication rounds T , client selection probability p_i , clients per round N_t , and local weights ω .
- 2: **for** $t \in [1, T]$ **do** ▷ communication rounds
- 3: $N_t \leftarrow \text{random.choice}(N, p_i)$ ▷ select subset of N with probability p_i per client
- 4: $\omega_{i \in N_t}^t \leftarrow \omega$
- 5: $\omega_{i \in N_t}^{t+1} \approx \min_{\omega'} \mathcal{L}(\omega'; \omega^{t+1})$
- 6: $\omega^{t+1} = \frac{1}{N} \sum_{i \in N_t} \omega_i^{t+1}$ ▷ aggregate parameters
- 7: **end for**

The last strategy, *MPIFed_{PROX}* (Li et al., 2018), addresses the issues derived from device heterogeneity. In this regard, full- and partial-straggler clients negatively impact training performance. This method proposes a proximal term to balance and ensure the stability and generalization of the global model. On this basis, local updates are constrained within the parameters of the initialized model, and the diverse features of heterogeneous hardware are adeptly managed, as demonstrated in Eq. (11). Here, κ represents the learning coefficient, and *SGD* denotes the local objective function. Then, each client sends ω_i^{t+1} to the server. The proposed FedProx scheme has been summarized as indicated in Algorithm 2.

4.3. Experimental settings

The experiments were conducted on the CETA-Ciemat supercomputer infrastructure. The used partition, known as Turgalium, is made up of 28 GPU-CPU nodes, where each node is built of CascadeLake 6263 CPUs running at 2.60 GHz from 192 to 768 GB RAM. Each node is equipped with one to four NVIDIA Volta V100 GPUs of 32 GB, 5120 CUDA cores, 640 Tensor Cores, and a memory bandwidth of 900 GB/s. The communication between GPUs is performed through a high-speed InfiniBand EDR100. A total of 10 executions have been conducted in each experiment.

The *Standalone Simulation* deploys a single process on a specific V100 GPU. In contrast, the *Distributed Computing* performs a parallel running by deploying 5 processes on 5 individual V100 GPUs. The case study is conducted using a changing number of clients. For the *Standalone Simulation*, the number of clients is set to $N = 20$ since the objective of the experiment is to provide a departure point for the consequent *Distributed Computing* experimentation. For the latter experiment, $N = 20$, $N = 50$, and $N = 100$ clients are launched to determine each algorithm classification. The batch size is set to $BS = 32$, the number of training epochs to $E = 10$, and the number

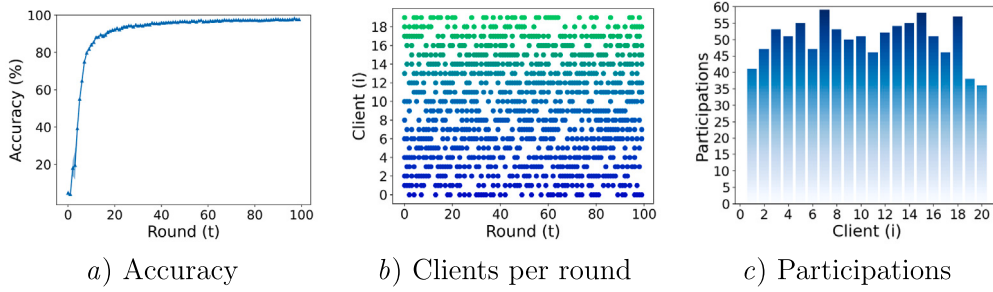


Fig. 5. Results obtained for the $SSFed_{DEC}$ algorithm with $N = 20$.

Table 5

Classification results obtained by the *Standalone Simulation* experiment. The last accuracy (LA) and maximum accuracy (MA) metrics are shown.

N	Metric	SPDec	SPFedAvg	SPHierarchical	SPVertical	SPTurbo
20	LA	97.72 ± 0.05	97.76 ± 0.11	92.51 ± 0.43	97.96 ± 0.04	95.77 ± 0.65
	MA	98.16 ± 0.01	97.81 ± 0.20	92.84 ± 0.66	97.96 ± 0.04	95.77 ± 0.65

of clients communicating per round is set to 10 for $N = 20$ (with a failure factor of 50%), and to 20 for both $N = 50$ and $N = 100$ (with failure factors of 60% and 80%, respectively). The number of communication rounds is adjusted to $T = 60$, $T = 100$, or $T = 200$, depending on the algorithm. Finally, an assessment of privacy concerns is conducted in light of *Byzantine attacks*, where a single client behaves maliciously. Additionally, an analysis is performed to understand the impact of decentralized data on training.

The training set comprises 80% of the dataset, leaving the remaining 20% for testing. The learning rate is set to 3×10^{-2} for both clients and servers, with a weight decay of 3×10^{-3} . SGD is selected as the most appropriate optimizer for a wide-spread perspective. The selected model is *ResNet50*.

4.4. Experimental discussion

4.4.1. Evaluation of standalone simulations

The results obtained in the first experiment are analyzed as follows. Initially, Fig. 5 presents the results of $SSFed_{DEC}$, with special emphasis on three distinctive facets: accuracy, client communications, and client participation.

Fig. 5(a) shows the accuracy evolution through the number of communication rounds. The improvement in accuracy becomes particularly evident in the initial rounds of communication, during which the model strives to generalize and capitalize on the information derived from diverse subsets of client data. Moreover, the model convergence has shown promising results, consistently advancing across all communication rounds and reaching peak accuracy. The negligible standard deviation demonstrates the robustness of the decentralized strategy across changing conditions during executions. Then, Fig. 5(b) depicts the clients involved in the communication rounds. As demonstrated, the simulation ensures that several clients do not share training information, thereby creating a realistic environment. Finally, Fig. 5(c) presents the aggregated number of client participations considering all communication rounds, highlighting varying degrees of collaboration. Factors such as computational or communication speed, fault tolerance, and other considerations influence this collaboration. In this experiment, 50% of the clients participated in each communication round. In this experiment, the decentralized approach $SPDec$ demonstrates superior performance. By updating each client before globally sharing the gradient, this method aims to mitigate the issue of uneven client participation in training during each iteration.

The Fig. 6 compares both types of experiments with the same number of clients $N = 20$. Fig. 6(a) shows the evolution of the evaluation accuracy for *Standalone Simulation* algorithms with $N = 20$.

Table 6

Classification results obtained by the *Distributed Computing* experiment. The last accuracy (LA) and maximum accuracy (MA) metrics are shown.

N	Metric	MPIFedAvg	MPIFedOpt	MPIFedProx
20	LA	97.66 ± 0.29	91.88 ± 0.60	96.72 ± 1.10
	MA	97.66 ± 0.29	92.46 ± 0.66	97.32 ± 0.43
50	LA	94.06 ± 1.98	90.88 ± 1.84	96.98 ± 0.94
	MA	97.03 ± 0.42	93.26 ± 1.01	97.99 ± 0.10
100	LA	95.80 ± 0.84	89.50 ± 0.93	95.64 ± 0.79
	MA	96.37 ± 0.44	89.50 ± 0.93	97.28 ± 0.04

The accuracy progression for these methods closely mirrors that of the decentralized method, yielding comparable values. The behavior of $SSFed_{HIE}$ diverges from the previous approaches. Although precision eventually converges to high levels, it falls short of achieving comparable performance metrics. This is primarily attributed to the limited number of clients engaged in the exchange of information and the formation of communication subgroups. Therefore, this method is employed in specific scenarios characterized by client communication failures that significantly hinder model convergence, showcasing resilience against straggling clients. Accuracy results are described in detail in Table 5 for all evaluated cases. The discrepancies among the top-3 methods are minimal, with a difference of only 0.25% accuracy between the highest performing strategy, $SSFed_{DEC}$, and the lowest, $SSFed_{AVG}$.

4.4.2. Delving into the distributed version

The second experiment yielded the following results. Initially, for $N = 20$, $MPIFed_{AVG}$ and $MPIFed_{PROX}$ exhibit similar behavior in terms of accuracy growth during the first communication rounds, as shown in Fig. 6(b). The obtained accuracy remains stable in the classification of the evaluation set. In contrast, the $MPIFed_{OPT}$ method requires more communications rounds to converge compared to the other methods. Nevertheless, the primary aim of this algorithm lies in ensuring data privacy and security during communication stages. Despite these outcomes, for $N = 50$ and $N = 100$, convergence requires a longer duration. In such instances, a significant drawback emerges. The heterogeneity of the distributed environment increases proportionally with the number of clients, thereby leading to challenges associated with heterogeneity. However, the benefits of incorporating more clients are noteworthy, particularly due to the increased volume of training data in a realistic training setting. As noted in the previous experiment, $MPIFed_{OPT}$ converges at a slower rate due to its emphasis on gradient compression and encryption to prioritize privacy over optimal performance.

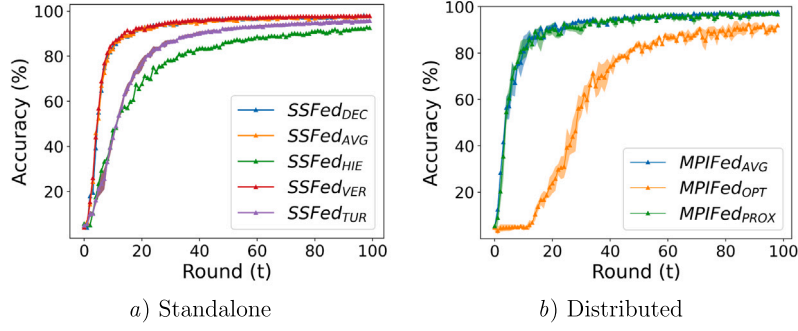


Fig. 6. Results obtained for the *Standalone Simulations* and *Distributed Computing* algorithm with $N = 20$.

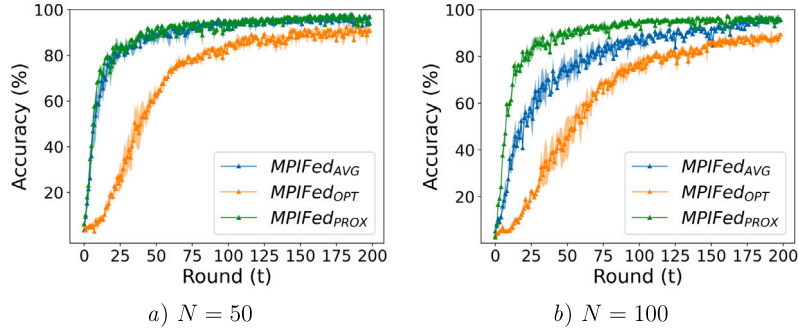
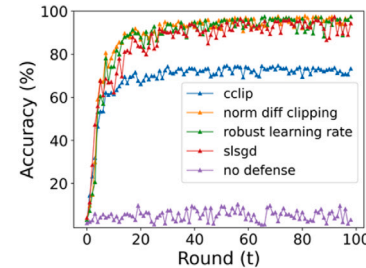


Fig. 7. Results obtained for the *Distributed Computing* algorithm with $N = 50$ and $N = 100$.

Table 7

Results of the effectiveness of different defense mechanisms against Byzantine attacks.

	Method	Last accuracy	Max accuracy
	cclip	73.25	74.69
	robust learning rate	97.42	97.46
	slsgd	94.39	97.36
	norm diff clipping	97.44	97.79
	no defense	3.18	10.26

The findings are highly promising, affirming the considerable potential of FL algorithms in effectively accommodating an expanding number of clients. The results are shown in Figs. 7(a,b) for the $MPiFed_{AVG}$, $MPiFed_{OPT}$, and $MPiFed_{PROX}$ methods with $N = 50$ and $N = 100$, respectively. Table 6 summarizes these results, demonstrating consistently high accuracy across the methods. However, the $MPiFed_{PROX}$ algorithm exhibits homogeneous behavior across all client distributions, whereas the convergence rate of $MPiFed_{AVG}$ deteriorates with an increase in the number of clients. In this context, $MPiFed_{PROX}$ ensures the stability and of local updates as the number of heterogeneous clients increases, simultaneously guaranteeing the generalization of the global model. This strategy achieves the highest accuracy, reaching 97.99% under the configuration of $N = 50$, which represents the optimal setting for the specific dataset and training conditions. On the other hand, the performance of $MPiFed_{OPT}$ is moderately affected by this increment in client distributions.

4.4.3. Integrity and privacy evaluation

Integrity and privacy are evaluated against Byzantine attacks. Various techniques have been examined, including centered clipping (*cclip*),

norm diff clipping, robust learning rate, and secure local stochastic gradient descent (*slsgd*) (Huang, Zhou, Wang, & Shi, 2021; Karimireddy, He, & Jaggi, 2020, 2021).

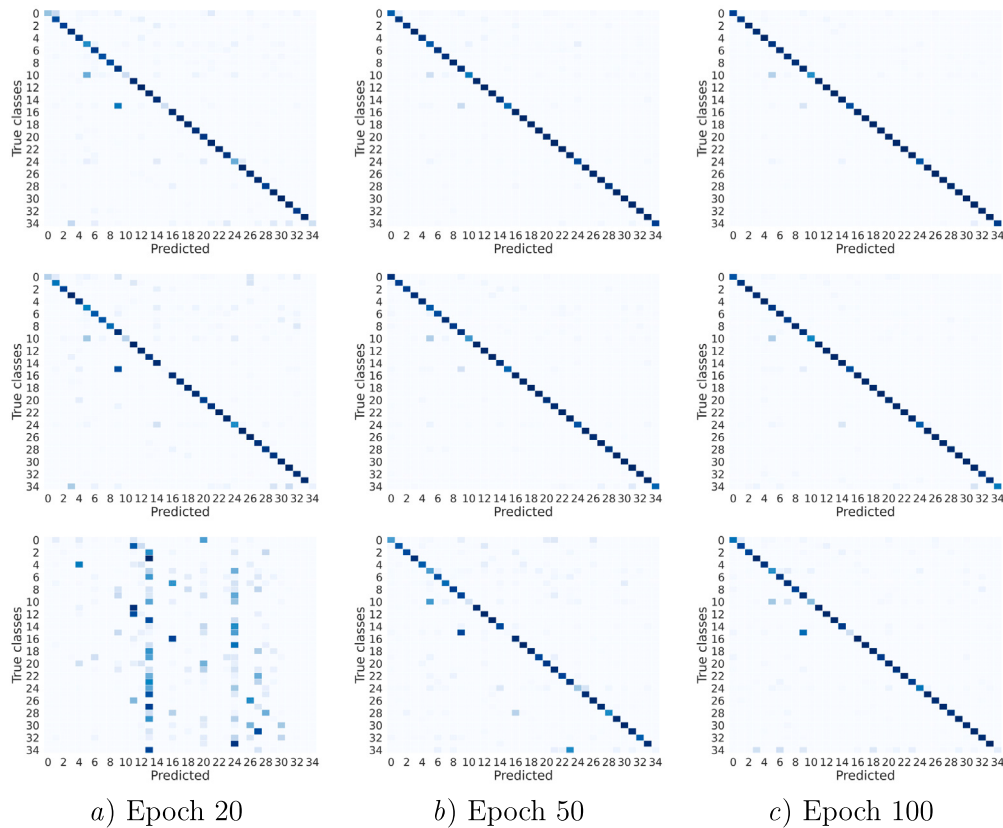
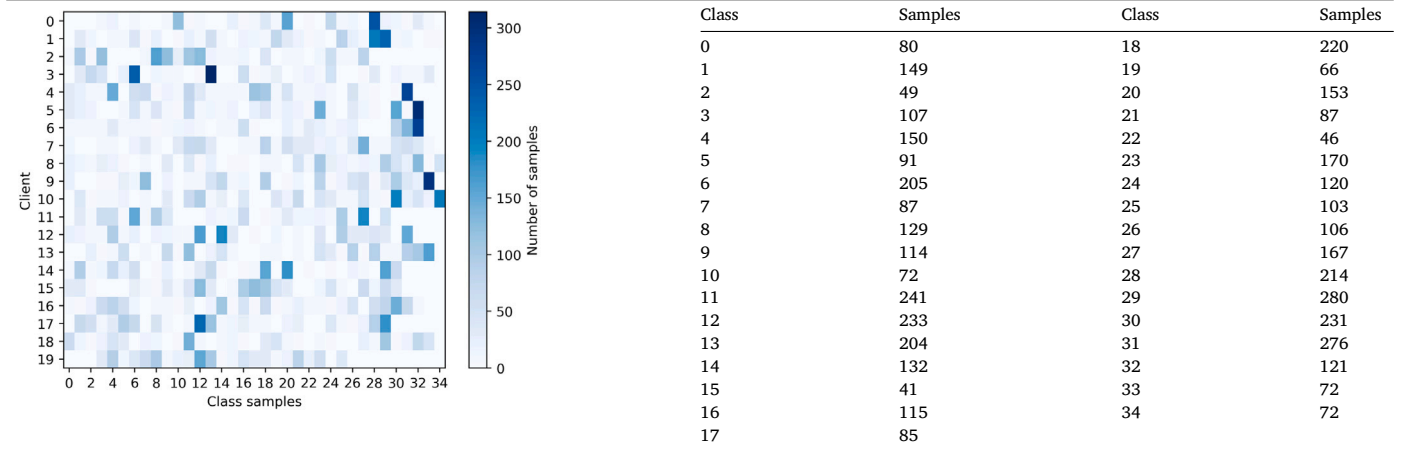
The objective is to evaluate the effectiveness of these methods compared to scenarios without defense mechanisms. The experiment is configured with $N = 20$, wherein one malicious client conducts attacks by injecting gradients with random values, thereby compromising the performance of the model. The accuracy of the model is depicted in Table 7. Notably, the defense methods showcase remarkable effectiveness. As observable, the defense techniques yield comparable accuracy values. However, *cclip* shows deviation from this trend. Nonetheless, FL is highly effective in dealing with intrusive Byzantine attacks.

4.4.4. Impact of decentralized data

The distribution of training samples offers valuable insights into the heterogeneous learning conditions encountered by individual clients, providing a comprehensive analysis of how these conditions influence model convergence. To ensure a consistent evaluation framework, the uniformity of test samples across all clients is intentionally maintained. This deliberate choice helps mitigate the influence of local training

Table 8

Sample distribution among classes for the train set (left heatmap) and the test set (right table).

**Fig. 8.** Classification confusion matrix for $N = 20$ obtained at different training epochs. The results for the *AVG*, *PROX* and *OPT* algorithms are organized in rows, respectively.

samples on each client model, fostering a more unbiased assessment of model generalization and enhancing the reliability of the comparative analysis. The data distribution is presented in Table 8. High variability is observed among the samples per class held by each client, with noticeable smaller classes predominating across the majority of clients, depicted in lighter colors.

The model performance across distinct classes is shown in the confusion matrices over multiple epochs, as illustrated in Fig. 8. These visual representations offer intuitive insights into the spatial distribution of predicted classes, providing a dynamic visualization of the model's adaptability throughout training. The inherent decentralization of data

sources contributes to creating diverse training landscapes for individual clients. Nevertheless, an examination of the confusion matrices throughout the epochs indicates that the model convergence is not compromised.

These classification maps serve as a crucial tool for scrutinizing the iterative evolution of the model. From this perspective, the intricate interplay between locally available training samples and the shared model information among clients is revealed. Concretely, classes such as *3-Transportation*, *9-Sapling*, and *24-Snow mountain* exhibit notable confusion during the initial training rounds. This information

enriches the understanding of the FL process, especially in terms of its adaptability to diverse RS datasets and collaborative learning dynamics.

5. Conclusions and remarks

This article introduces a comprehensive study of novel FL methodologies, with a specific focus on evaluating the classification outcomes of crowd-sourced high-resolution optical RS scenes. For this purpose, this study uses the FedML library to enable the integration of RS data imagery with a wide range of novel methodologies. Two alternatives are under evaluation: simulations and real-world scenarios through distributed computing. In this context, the research is motivated by the trend towards secure and privacy-aware crowd-sourced decentralized training. In such scenarios, training a global classification model involves a large number of clients contributing location-dependent information for collaborative learning without compromising proprietary data in a scalable environment. To this end, crowd-sourced data is the ideal case study, providing a real-world scenario with disjoint information.

The presented study demonstrates the generalization ability of conventional models under the FL paradigm using high-performance DNNs. The study reports on the volume of communications performed by clients, demonstrating the effectiveness of handling delayed clients under changing conditions. Across various experiments, the results highlight robust performance in processing RS datasets for each method. This finding opens the door for a promising convergence between the FL and RS domains in the near future. Additionally, the results demonstrate remarkable resilience against Byzantine attacks, underscoring its robust capabilities in mitigating the impact of malicious clients. Therefore, these findings yield positive insights in scenarios characterized by limited labeled samples and data imbalances, where performance is often hindered by heterogeneity issues. This concern is mitigated within decentralized federated environments, and it becomes more pronounced when dealing with remotely sensed data due to its limited spatial coverage and associated security considerations.

Finally, this work identifies several challenges that require future research. These include the computational requirements of each device, potentially resulting in clients with outdated parameters participating infrequently in training. Indeed, data heterogeneity requires applying preprocessing techniques such as data normalization and standardization, among other approaches. As a summary, this work lays the groundwork for building robust methodologies in RS applications, taking advantage of richer coverage of the surface by collecting scenes from multiple locations in a secure environment. Also, the conducted case of study offers future promising solutions for various types of imagery, including hyperspectral scenes. This could determine meaningful insights into ground materials by analyzing spectral signatures across disjoint captured areas, thereby enhancing performance.

CRedit authorship contribution statement

Sergio Moreno-Álvarez: Conceptualization, Methodology, Software, Writing. **Mercedes E. Paoletti:** Writing – original draft. **Andres J. Sanchez-Fernandez:** Visualization, Investigation, Writing. **Juan A. Rico-Gallego:** Supervision. **Lirong Han:** Conceptualization, Methodology, Software, Writing – review. **Juan M. Haut:** Editing, Supervision.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Sergio Moreno Alvarez reports administrative support and equipment, drugs, or supplies were provided by Center for Advanced Technologies Extremadura. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

We provide source code in GitHub.

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